DEEP RESEARCH PROMPT — MISSION CRITICAL SOFTWARE & CYBERSECURITY

1. MACRO DRIVERS
   - Which macroeconomic variables most directly affect NRR, churn, and new ARR growth for mission-critical software and cybersecurity companies? Rank by historical explanatory power where data exists.
   - How does the current macro environment (rates, USD, enterprise capex cycle as of Q1 2026) compare to prior periods when this cohort re-rated?
   - Is there a documented relationship between IT budget growth and mission-critical software outperformance vs. the broader software index?
2. HISTORICAL ANALOGUES
   - Has a comparable "indiscriminate sell-off driven by perceived technology substitution risk" occurred before in software equities?
   - Specifically: how did mission-critical software trade during (a) the cloud transition scare for on-premise vendors (2010–2014), (b) the SaaS dislocation of 2022, and (c) any prior period where a technology wave was feared to obsolete incumbent software?
   - What were the return profiles and recovery timelines in those analogues? What were the false recoveries vs. sustained re-ratings?
3. TAM & GROWTH RATE
   - What is the current estimated TAM for: (a) mission-critical / vertical market software globally, (b) enterprise cybersecurity software globally?
   - What are the consensus CAGR estimates for 2025–2030 from Gartner, IDC, McKinsey, or equivalent? Note source and vintage of estimate.
   - How does LLM adoption change the TAM calculus — is it net additive (AI-augmented workflows expand spend) or net subtractive (replacement)?
4. DISRUPTION RISK
   - What would a realistic LLM disruption scenario look like for this theme in 2–5 years? Which sub-verticals are most vulnerable: ITSM, security, ERP, clinical, financial infrastructure?
   - Are there any venture-backed or hyperscaler-native competitors building LLM-native replacements for mission-critical workflows? What is their current traction (ARR, customer count, funding)?

- What structural barriers (regulatory approval, liability, data sovereignty, integration complexity) would slow substitution even if LLM capability were sufficient?

5. CATALYST CALENDAR
   - List all known upcoming events in the next 6–12 months that could materially move this theme: earnings dates for key names, regulatory decision dates (NIS2 enforcement, DORA go-live, AI Act phase-in), major IT spending surveys (Gartner, IDC), Fed meetings, and any sector-specific conferences (RSA, Microsoft Ignite, Salesforce Dreamforce).
   - Flag which catalysts are binary (pass/fail, beat/miss) vs. directional.
   - Identify the single most important data point to watch in the next 90 days that would either confirm or invalidate the NRR resilience thesis.

Mission-critical software and cybersecurity fundamentals are still primarily driven by classic enterprise IT and macro cycles, but the current AI shock has added an overlay of "terminal value risk" that is historically unusual and is driving the indiscriminate re-rating you're reacting to.[1] [2]

1. Macro drivers for NRR, churn, new ARR

Key macro variables (ranked by historical explanatory power)

For mission-critical software and cybersecurity, the variables that most consistently explain variation in NRR/churn/new ARR over cycles have been:

1. CIO / IT budget growth and enterprise software spending growth
   - Directly links to expansion ARR (seat growth, module cross-sell) and new logo formation. In slowdowns (2022–23), broad SaaS growth decelerated sharply as customers reassessed and cut SaaS spend, and retention weakened for non-mission-critical tools.[3] [4]
   - Gartner-style forecasts of IT and software spending are typically the single best top-down predictor of sector revenue growth, especially for new ARR.
2. Corporate profit growth and enterprise capex cycles
   - When core IT spending accelerates (e.g., Gartner is now projecting 12.5 percent growth in core IT spending in 2025 after strong 2024, with particularly sharp upswing in datacenter systems), software and infra budgets follow, supporting upsell and new projects.[5]
   - In profit squeezes, CIOs prioritize "keep the lights on" and security; this tends to protect mission-critical/secure workloads but hurts discretionary SaaS.
3. Interest rates / discount rates
   - Rates work mostly through equity multiples and customer behavior rather than directly through NRR. Higher rates in 2022–23 drove large SaaS multiple compression even as top-quartile public SaaS still grew high-30s/40s percent, but also pushed

customers toward cost optimization and vendor consolidation, which pressured marginal tools more than core systems.[4]

4. FX (USD strength)
    - FX matters for reported NRR and ARR in global names with high non-USD exposure; dollar strength in 2022–23 was a measurable headwind to reported growth for US-listed software, including security, but rarely changed churn qualitatively.
    - For European vertical-software and cybersecurity vendors, EUR/CHF moves affect reported growth and margins but are second-order vs local IT budgets.

Roughly, CIO/IT and software spending growth plus corporate profit trends explain most of the *fundamental* variation in expansion ARR and new ARR; rates and FX explain a large part of the *valuation* variation.

Current macro vs prior re-rating episodes (Q1 2026 snapshot)

- IT and software spending:
    - Global IT spending is projected to top about 6 trillion dollars in 2026, nearly ten percent higher year-over-year, with security, AI and cloud infrastructure as primary drivers.[6]
    - The broader enterprise software market (including security, databases, infra) was about 899.9 billion dollars in 2024, growing toward 2025–30 at roughly mid-single-digit CAGR; cybersecurity spend in this mix is growing materially faster (mid-teens).[3]
    - This is a very different backdrop from 2022, when SaaS growth fell to the slowest in several years and more than half of SaaS businesses saw lower retention rates, reflecting broad budget tightening.[4]
- Rates and discount rates:
    - Policy rates have moved down from their 2023/early-24 peaks but remain well above the ZIRP/QE regime that underpinned 2019–21 SaaS multiples; expect structurally higher cost of capital vs 2010–21, which caps the "nosebleed multiple" regime even if growth re-accelerates.
    - Compared with the 2010–14 cloud transition, we now have both higher absolute rates and higher term premia, which suggests a lower structural P/S ceiling for growth software than in prior cycles.
- Enterprise capex and AI:
    - Gartner and other forecasters see unusually strong datacenter systems spending for 2025 (nearly 47 percent growth to about 489.5 billion dollars as AI workloads drive infra upgrades), and continued robust core IT spending growth.[5]
    - This favors infra, platform, and security vendors more than traditional seat-based horizontal SaaS, but vertical and mission-critical systems that tie into AI workflows should benefit.

IT budget growth vs software index outperformance

- There is a long-documented relationship between periods of above-trend IT spending and software sector outperformance vs broader indices. Gartner projections show core IT spending growing low double-digits into 2025, while enterprise software and cybersecurity subsets grow faster than overall IT, reflecting mix-shift toward software.[3] [5]
- Public SaaS data during 2020–21 illustrate the linkage: in an environment of ultra-loose macro policy and aggressive digital transformation, top decile public SaaS companies grew 66 percent in 2021, dropping to 48 percent in 2022 as macro tightened; sector indices outperformed strongly during the high IT-spend phase and underperformed during the slowdown.[4]
- Within that, security and mission-critical vertical software tend to outperform broader software indices particularly during stress periods because budget priorities favor resilience, regulatory compliance and operational systems of record over discretionary apps.[3]

2. Historical analogues: substitution scares and recovery

Indiscriminate substitution sell-offs

The current "AI agent breaks per-seat SaaS" narrative has analogues but is unusually broad:

- 2010–14 cloud transition (on-prem to SaaS):
  - Fears that cloud SaaS would obsolete on-premise vendors (especially license-plus-maintenance models) led to a multi-year value rotation within software.
  - Many on-prem vendors derated versus high-growth SaaS, but mission-critical on-prem names with strong maintenance revenue and gradual cloud migration paths (e.g., vertical software, ERP incumbents) ultimately preserved value by offering cloud-adjacent products rather than being displaced quickly.[7]
- 2022 SaaS dislocation (rates + macro + "too many tools"):
  - Top quartile SaaS growth decelerated sharply and retention fell for more than half of SaaS businesses as customers rationalized spend.[4]
  - Sector drawdowns were severe; yet, mission-critical and security names generally held up better in fundamentals (smaller churn uptick, more stable NRR) even as their multiples compressed alongside the group.
- 2024–26 AI / agent shock:
  - The recent episode has been framed explicitly as a structural threat to the per-seat SaaS model; one analysis estimates that Anthropic's Claude Cowork launch triggered about 285 billion dollars in software market cap losses over 48 hours and highlighted a "390 billion dollar per-seat industry" whose economics are now in question.[2]
  - This is closer to a "terminal value shock" than prior cycles, because it targets the core pricing unit rather than deployment model or budget level.

How mission-critical names traded in these episodes

- 2010–14 cloud scare:
  - Core ERP, financials, and vertical systems with high switching costs typically saw muted revenue impact, with maintenance and support streams remaining resilient.
  - Valuations lagged fast-growing SaaS during the euphoria phase, then outperformed as initial expectations of rapid obsolescence proved overstated.
- 2022 SaaS dislocation:
  - Growth rates slowed broadly, but top decile public SaaS still grew 48 percent in 2022, top quartile 39 percent, and mission-critical names often maintained high-teens to 20s percent growth alongside very high NRR.[4]
  - Recovery timelines: many quality SaaS names bottomed between late 2022 and H1 2023 and then saw a partial re-rating as macro stabilized; weaker, non-essential tools saw false dawns where brief rallies faded due to continued churn and down-sell.
- Current AI-agent shock:
  - Mission-critical systems of record, vertical market software, and core security control-points are again seeing large multiple compression despite limited evidence of near-term churn; early data show that installed bases remain intact while pricing models and packaging are being re-negotiated.[1] [2]
  - Recovery paths from prior episodes suggest:
    - False recoveries occur when rallies are driven by sentiment or rates but fundamentals (NRR, new ARR, pricing power) continue to erode.
    - Sustained re-ratings follow clear evidence that: (a) IT/security budgets remain robust, (b) NRR stabilizes, and (c) incumbents successfully **monetize** the new tech wave (cloud then, AI now) rather than being displaced.

Timelines in prior scares were typically 12–24 months from peak fear to sustained re-rating for the high-quality cohort, with several quarter-long bear-market rallies along the way.

3. TAM and growth rates (2025–2030)

TAM: mission-critical / vertical software

- A recent market report puts the vertical software market at about 146.68 billion dollars in 2025, projected to reach 254.82 billion dollars by 2030, implying 11.68 percent CAGR.[8]
- The broader enterprise software market, using Gartner's definition (including infra, security, databases), was about 899.9 billion dollars in 2024.[3]
- Mission-critical and vertical software can be thought of as a high-value subset of this, likely in the low- to mid-hundreds of billions of dollars today; the cited vertical software figure is a reasonable proxy for explicitly vertical, workflow-embedded systems.[8] [3]

TAM: enterprise cybersecurity software

- The enterprise software market overview above notes that cybersecurity spending is projected to increase about 15 percent in 2025, faster than overall enterprise software growth.[3]
- Independent security spending estimates from major analysts (Gartner and peers referenced in that piece) put security software and services on a trajectory of mid-teens growth through the second half of the decade, implying a TAM growing from low-hundreds of billions of dollars today to potentially double that by 2030.[3]

Consensus growth (Gartner / IDC / McKinsey-style)

- Enterprise software (broad):
  - One synthesis of Gartner projections for 2025–30 suggests around 4.99 percent CAGR for enterprise software revenue, reaching roughly 403.90 billion dollars by 2030 on a narrower definition, while the broader category including infra and security is larger and growing somewhat faster due to mix.[3]
- Vertical software:
  - The vertical software report explicitly states an 11.68 percent CAGR from 2025 to 2030, from about 146.68 billion to 254.82 billion dollars.[8]
- Cybersecurity:
  - Cybersecurity spending inside IT is projected to increase 15 percent in 2025, with forward commentary indicating sustained double-digit growth through 2030 as regulatory pressure and AI-driven threat complexity intensify.[3]

These figures align with McKinsey's broader TMT view that AI and software are core growth engines for European and global TMT value pools over the next decade, with AI-intensive software and security as "critical battlegrounds."[9]

LLM adoption: net additive vs subtractive TAM

- Net additive channels:
  - New AI-augmented workflows expand addressable processes (e.g., agentic support, AI-driven analytics, code security, automated incident response), demanding new modules and upgrades within mission-critical systems and security platforms.
  - There is already evidence that organizations are shifting from exploratory AI pilots to production-scale deployments requiring stronger governance, infrastructure and cost control, which drives spend on secure, governed platforms.[6]
  - Vertical software TAM could expand as industry-specific AI features, copilots and compliance tooling command premium ARPU within tightly regulated domains.[8]
- Net subtractive / compression channels:
  - Per-seat pricing models for knowledge-worker SaaS are under direct pressure as AI agents can do the work of multiple human "seats," weakening the link between user count and value.[2]

- Workflow-generic tools (basic productivity, point-solution SaaS) are particularly exposed to replacement or consolidation into broader platforms.

Net, TAM for mission-critical and security *likely expands* (more workloads, higher value per workflow) even as unit economics change (e.g., higher value per account but fewer seats, or more usage-based/pricing tied to outcomes). The net effect for vertical market software and security looks additive to the top-line opportunity but mixed for specific seat-based pricing models.[6] [2] [8]

4. LLM disruption risk and barriers

Realistic 2–5 year disruption scenarios

Over the next 2–5 years, realistic disruption for mission-critical software and cybersecurity is more about **pricing and surface area** than wholesale system replacement:

- ITSM / DevOps / observability:
  - High vulnerability to agentic automation reducing "seat" counts (fewer human operators, more automated triage) but strong opportunity for incumbents to embed agents into existing platforms.
  - Agents can automate ticket routing, remediation and incident response; incumbents that fail to ship credible AI workflows risk share loss to LLM-native ops platforms.
- Security (SOC, EDR, identity, SASE):
  - SOC workflows are highly suited to LLMs (triage, summarization, playbook generation).
  - Disruption risk is real at the level of specific tools (e.g., rule-based SIEM, basic alerting) but incumbents with data moats and regulatory trust can turn LLMs into force multipliers rather than cannibalizers.
- ERP / financial and HR systems:
  - Core ledgers and compliance workflows are unlikely to be replaced in 2–5 years; instead, copilots and agents will sit on top of ERP/HCM data to automate reporting, approvals and forecasting.
  - Disruption risk lies more in new "AI front-ends" that intermediate user interactions and may weaken the visible lock-in of incumbents if they become commoditized back-ends.
- Clinical / healthcare:
  - Strong potential for AI agents in documentation, coding, prior auth and decision support, but regulatory constraints, liability and need for explainability heavily favor incumbents and specialized vertical players over green-field LLM tools.
- Financial infrastructure:
  - LLMs will automate operations (reconciliations, exception handling, compliance checks) and customer interactions, but core transaction systems and risk engines are unlikely to be displaced quickly.

LLM-native challengers and hyperscaler moves

- There is a growing set of LLM-native SaaS and security startups explicitly targeting existing workflows, especially in:
  - AI-native customer support and ITSM,
  - AI-driven application security and code scanning,
  - AI copilots for ERP, CRM and vertical systems.
- Many of these are early-stage with limited ARR and customer counts; they often focus on narrow workflows and rely heavily on hyperscaler or foundation model APIs for core capability.[10]
- Hyperscalers (Microsoft, Google, AWS) are embedding LLMs and agents into their cloud platforms, security suites, and productivity/ERP stacks, effectively creating "LLM-native" extensions that can cannibalize some independent tools while reinforcing their own ecosystems.[9] [6]

At this stage, documented ARR and customer metrics for individual venture-backed LLM-native vertical vendors are sparse in public sources; where disclosed, they are still tiny relative to incumbents.

Structural barriers slowing substitution

Even with rapid LLM capability gains, several barriers structurally slow direct substitution of mission-critical systems:

- Regulatory and compliance hurdles
  - In Europe, NIS2 has now fully transitioned into national law and as of 2026 is under active enforcement; organizations face audits, documentation demands and fines for non-compliance.[11]
  - Security and critical infrastructure software must meet stringent standards, making it hard for unproven LLM-native tools to displace incumbents quickly.
- Liability and risk management
  - In clinical, financial and critical infrastructure workflows, errors from AI systems carry heavy legal and financial liability; enterprises prefer vendors with strong guarantees, audit trails and established track records.
- Data sovereignty and residency
  - Many sectors require data to reside in specific jurisdictions or under specific controls; integrating LLMs into these environments often demands complex architectures and vendor certifications.
- Integration complexity and switching costs
  - Mission-critical systems are deeply integrated into dozens of downstream processes and databases; replacing them is a multi-year program with high execution risk.
  - LLM-native tools often sit as overlays or assistants rather than full system replacements, at least in the 2–5 year horizon.

These barriers strongly favor incumbents that can **integrate** LLMs into existing products while maintaining compliance and governance.

5. Catalyst calendar (6–12 months) and key data point

Regulatory and structural catalysts (EU-centric)

- NIS2 enforcement (EU):
  - Transposition deadline passed on 17 October 2024; as of 2026, NIS2 obligations are fully in force and national authorities are actively auditing and enforcing compliance.[11]
  - Ongoing implementing acts and guidance from the European Commission will continue through 2026; each technical clarification can drive incremental security spend.[11]
  - Nature: mostly directional (more/less regulatory pressure), not binary.
- DORA (Digital Operational Resilience Act, EU financial sector):
  - Key go-live date is January 2025; 2026 is about enforcement ramp-up, supervisory expectations and potential guidance updates, which should sustain demand for operational resilience, monitoring and security tooling.
  - Nature: directional tailwind for financial infrastructure and security.
- EU AI Act phase-in:
  - The AI Act is entering phased implementation over 2025–27, with high-risk use-case obligations being defined and enforced.
  - Nature: directional; increases demand for AI governance, monitoring, logging and risk-management capabilities in software and security stacks.

IT spending and survey catalysts

- Major IT spending surveys and outlooks (Gartner, IDC and others):
  - Gartner has already raised core IT and datacenter systems spending forecasts for 2025 and issued 2026 predictions, highlighting strong infra and AI-related spend.[5]
  - Additional mid-year 2026 updates and CIO surveys on AI/agent adoption, security and software budgets will be important directional catalysts for the theme.
  - Nature: directional (IT budget upgrades/downgrades, AI adoption sentiment).

Macro: Fed meetings and rates

- FOMC meetings through 2026 (roughly every 6–8 weeks) will remain important for discount rates and therefore valuations.
- Nature: mostly directional, with occasional binary feel when cuts/holds vs expectations are at stake.

Sector conferences (directional)

Key conferences likely to matter over the next 6–12 months include:

- RSA Conference 2026 (security):
  - Major venue for security vendors to showcase AI-driven products, agentic SOC capabilities, and regulatory-driven offerings; tone on AI disruption vs opportunity in security will affect sentiment.
  - Nature: directional.
- Microsoft Ignite, Microsoft Build, Google Cloud Next, AWS re:Invent:
  - Hyperscaler AI and security product news; agents/platform announcements can either validate or undercut specific sub-verticals (e.g., ITSM, observability, security).
  - Nature: directional.
- Dreamforce (Salesforce) and other large SaaS user conferences:
  - Updates on AI monetization (copilot attach, agent usage, pricing), NRR trends and seat dynamics in large horizontal SaaS.
  - Nature: directional but sometimes binary at the single-name level (beat/miss on AI expansion).

Earnings: binary vs directional

For the theme, upcoming earnings cycles (Q1–Q4 2026) for:

- Mission-critical vertical software names
- Core ERP/HCM vendors
- Major cybersecurity platforms
- Hyperscaler cloud/security franchises

will provide the clearest data on:

- NRR and churn in AI-exposed vs mission-critical cohorts
- Seat counts vs agent usage trends
- AI feature attach rates and ARPU uplift
- Security and compliance-driven demand under NIS2/DORA/AI Act

At the single-company level, earnings are binary (beat/miss) catalysts; at theme level, several consecutive quarters will create a directional picture.

Single most important data point (next 90 days)

Over the next 90 days, the single most important data point for your NRR-resilience thesis is:

> Reported NRR (and related expansion metrics) for a representative basket of mission-critical vertical software and cybersecurity platforms in the upcoming earnings season, specifically segmented commentary on seat counts vs AI-agent usage and AI feature monetization.

If, across several core names, you see:

- NRR remaining very high (e.g., stable or modestly lower vs pre-AI),
- Seat counts being managed but offset by higher ARPU from AI features and deeper workflow penetration, and
- No meaningful evidence of agent-driven churn in core systems,

then the thesis that "mission-critical NRR is structurally resilient and AI is more monetization opportunity than terminal risk" is confirmed. If instead:

- NRR drops materially,
- Management reports visible seat compression not compensated by AI monetization, and
- Security/vertical vendors start to see real churn or pricing pressure,

then the thesis is challenged and the sector's re-rating path likely stretches out or shifts toward a narrower subset of winners.[1] [2] [4]

If you share your current mission-critical basket (tickers and regions), I can translate this into a concrete 6–8 KPI "earnings tracker" template you can hand to a junior or embed in the CAPE Theme Engine.

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